Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.
Digital Object Identifier 10.1109/ACCESS.2026.DOI

Temporal Pixel Masking for Reducing OCR Recovery from Short-Exposure Screen Photographs: A Single-UVC-Camera Feasibility Study

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[TODO: Funding acknowledgment to be filled.] This work includes a planned minimal-risk human-subject study. The authors' institution does not maintain an ethics review board for behavioral studies of this type; the study protocol follows the informed-consent and safety procedures described in Section V-D.

ABSTRACT Screen content can be silently captured by smartphones or smart glasses and fed to optical character recognition (OCR) or vision-language models (VLMs) for text extraction. We propose a temporal pixel mask that exploits the gap between human temporal integration and short camera exposures: a CSPRNG-driven per-cycle randomization decomposes each frame into rapidly alternating complementary subframes, so that a human observer perceives the complete image while a short-exposure camera records only a sparse pixel subset. In a feasibility study with a single USB Video Class (UVC) webcam spanning 9 distance/angle configurations and 10,575 captures, best-of-engine OCR character recovery under short exposure dropped from 94.1% (unprotected) to 15.1% for the readability-priority deployed profile and 5.0% (CI [4.3, 5.8]%, approaching but not conclusively meeting the $<5\%$ target) for the capture-hardened profile, with exact-match rates falling from 65.7% to 0.4% and 0%, respectively. The method's security boundary is explicit: video temporal averaging ($\geq$67 ms recording) recovers 47.9–71.1% character; three commercial VLMs recover up to 77.8% exact-match on large-font short text at 0.5 m; and under long exposure, temporal masking can paradoxically *increase* recovery relative to unprotected baselines by reducing effective luminance into the sensor's linear range. These boundaries confine the contribution to conventional OCR short-exposure mitigation within the tested UVC link. Actual readability and visual comfort await the planned user study.

INDEX TERMS Adversarial perturbation, anti-photography, camera capture, display privacy, OCR, persistence of vision, temporal masking, visual eavesdropping

I. INTRODUCTION

SCREENS are ubiquitous in offices, banks, hospitals, and conference rooms, and so are capture devices: smartphones, smart glasses (e.g., Ray-Ban Meta) [1], and miniature cameras. A single photograph of a screen suffices to extract text via OCR or identify interface elements through object detection. This *visual eavesdropping* is covert: it leaves no network traces and requires no physical contact with the target device. Controlled experiments and field studies confirm that such attacks are common, succeed at high rates, and frequently go unnoticed [2], [3]. Although these devices motivate our work, the physical evidence presented here covers only a single UVC webcam; reproducibility on smartphones, smart glasses, and other ISP pipelines is left to future work.

A. EXISTING APPROACHES AND THEIR LIMITATIONS

Current screen-privacy solutions each occupy a limited niche. Physical privacy filters restrict viewing angles but cannot stop a camera aimed squarely at the display. Software-based degradation (dimming, blurring, pixelation) sacrifices readability or depends on viewing geometry, and none specifically blocks on-axis capture (quantitative comparisons appear in §V-F). Digital watermarking provides post-hoc accountability rather than pre-emptive blocking [4]. Temporal re-encoding schemes designed for cinema anti-piracy (e.g., Kaleido [5]) target continuous video recordings and do not address short-exposure single-frame capture of static sensitive content or machine recognition (§II). These approaches cover viewing-angle restrictions, post-hoc tracing, and continuous-recording disruption; on-axis short-exposure capture followed by machine recognition falls in the gap between them.

The difficulty is that degrading information for the camera also tends to degrade it for the legitimate viewer. We ask whether display temporal structure can widen this gap under short-exposure capture.

B. CORE IDEA

Our approach exploits the gap between human temporal integration and camera exposure windows. Each frame is decomposed into n randomly complementary subframes that alternate at high speed. When the exposure is shorter than or close to one subframe duration, the camera records only an incomplete pixel subset, whereas the human observer integrates content across subframes (Fig. 1). Because camera exposure is not instantaneous sampling, long exposure or video reconstruction covering a full cycle can re-sum the complementary subframes; we therefore make no irrecoverability guarantee for arbitrary exposure conditions.

We model short exposure as a relevant first-tier attack: an observer snaps one brief photograph without calibrating exposure or holding the camera on the screen for multiple frames. We do not claim that every smartphone defaults to a subframe-length shutter; the S600 experiments use fixed, calibrated exposures rather than smartphone auto-exposure. Critically, the calibrated fixed-exposure protocol is *more favorable to the attacker* than typical smartphone auto-exposure: auto-exposure on a bright display in indoor lighting often selects short shutter speeds ($\leq$2 ms) that would capture less than one subframe, whereas our 3.91 ms calibration captures nearly a full slot. The experimental condition thus represents a stronger, not weaker, attacker within the short-exposure tier. A capable attacker can deliberately extend integration or record video; those modes form explicit higher tiers and a security boundary rather than an excluded case. Beyond exposure-based attacks, modern VLMs bypass the binarization bottleneck entirely: even from a single short-exposure photograph, language priors can complete partial glyph fragments, recovering up to 77.8% exact-match at 0.5 m (§V-C). This confines the practical contribution of temporal masking to conventional OCR short-exposure mitigation.

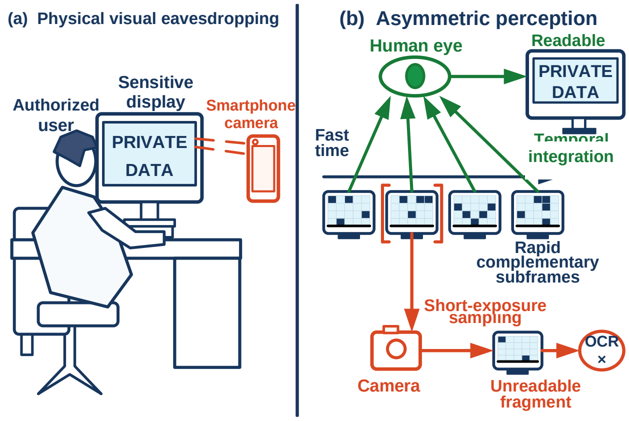

FIGURE 1. Threat scenario and operating boundary of temporal pixel masking. (a) A nearby camera can capture sensitive on-screen content without accessing the host system. (b) Human vision integrates complementary subframes, while a short camera exposure can intersect only part of a display cycle and leave OCR with an incomplete fragment. The camera path is conceptual: it does not assume zero-duration sampling or a fixed human integration time, and long exposure or multi-frame reconstruction can recover much more information.

C. CONTRIBUTIONS

The main contributions of this paper are:

1) **A temporal pixel mask implementation targeting conventional OCR under short-exposure capture**: Building on prior temporal masking ideas, we implement CSPRNG-driven pixel-level complementary subframes with profile-dependent stripe/glyph overlays and a GPU rendering prototype. An optional complementary adversarial noise layer yields negative net benefit under real-capture short exposure (25.6% vs. mask-only 19.2%) but is retained in enhanced profiles for its integration-attack and VLM-disruption contributions (§IV); the mask-only baseline is the cleaner reference for evaluating short-exposure performance in isolation. Effectiveness depends on camera exposure windows and the attacker's recognition pipeline.
2) **Multi-geometry real-capture OCR evaluation on a single UVC camera**: Using one eMeet SmartCam S600 USB webcam, we aggregate 10,575 captures across 3 distances and 3 angles. The corpus comprises component ablations, parameter sweeps, and attack evaluations rather than a balanced full-factorial design; validity conclusions draw from the core short-exposure OCR subset: 15.1% character recovery (0.4% exact) for the deployed profile and 5.0% (0% exact) for the capture-hardened profile.
3) **Cross-task stress tests and VLM boundary probes**: Beyond OCR, we report results for 4 object detectors on COCO val2017 (150 real-capture images, plus 8 for simulation diagnostics), MOT17 detection/association (5,316 simulated frames and 450 real-capture still frames), and 3 commercial VLMs at 0° on-axis across three distances. The VLM experiment is negative boundary evidence: conventional OCR protection does not transfer to modern end-to-end vision recognition.
4) **Security-frontier analysis**: We quantify residual leakage under integration attacks: ideal full-cycle integration recovers 95.2% in digital simulation; real-capture video temporal averaging against the deployed profile still recovers 71.1% char. The capture-hardened profile reduces video leakage to 47.9% char / 5.6% exact, but this attack class remains the fundamental boundary of the scheme. VLM probes further show that short-exposure protection fails for large-font short snippets: at 0.5 m Qwen3-VL reaches 77.8% exact, and at 1.5 m

the three models retain 33.3–47.2% exact. In the VLM era, the primary value of this analysis is not the defense itself but the empirical characterization of *where and why* temporal masking fails: content type, font size, and attacker capability jointly determine the protection boundary, providing actionable design constraints for next-generation display-privacy systems.

5) **Security–usability proxy analysis**: We report FPI, ΔE, SSIM, and normalized mutual information as exploratory digital proxies with Pareto fronts across 12 (n, f_r) configurations, distinguishing digital integration proxies from actual user experience. These proxies enable cross-configuration comparison but do not constitute validated perceptual thresholds; a controlled within-subject protocol is specified for subsequent usability validation.

The remainder of this paper is organized as follows. Section II reviews prior display-privacy and temporal-modulation work. Section III defines attacker tiers and protection goals. Section IV presents the masking design, and Section V reports OCR, VLM, cross-task, usability-protocol, and security-frontier evaluations. Sections VI and VII discuss the deployment boundary and conclude the paper.

II. RELATED WORK

A. SCREEN VISUAL EAVESDROPPING THREATS

Remote screen reading attacks (telescopes, telephoto lenses) [6], deep-learning reconstruction of HDMI electromagnetic leakage [7], covert capture by smart glasses [1], and indirect eavesdropping through reflections and refractions [8] have all been documented. A controlled experiment by the Ponemon Institute found that 91% of visual hacking attempts succeed within 15 minutes [2]. Eiband et al. [3] surveyed 174 participants on the prevalence of shoulder surfing and the coping strategies users adopt. Tang and Shin [9] proposed Eye-Shield, which uses the front-facing camera to detect bystanders and blur sensitive regions in real time, though it relies on user-side hardware and provides only localized masking. Our work focuses on display-side mitigation that requires no attacker-side cooperation, without claiming prevention of information acquisition under all exposure and reconstruction conditions.

B. DISPLAY PRIVACY PROTECTION

Physical privacy filters restrict viewing angles but cannot block an on-axis camera. Chen et al. proposed HideScreen, a display-side shoulder-surfing defense that suppresses low-frequency visual cues so screen content blends with the background beyond a designed viewing distance [10]; it targets human shoulder surfing in the spatial domain rather than short-exposure camera capture or machine recognition. Zhu et al. [11] proposed LiShield, which modulates smart LED illumination to interfere with rolling-shutter cameras, but the scheme acts on ambient lighting rather than displayed content. Visual cryptography splits a secret image into shares that combine upon overlay [12], inspiring temporal decomposition ideas, though its original formulation targets printed media. Digital watermarking [13] can embed tracking identifiers, including screen-shooting-resilient variants [4], but provides post-hoc attribution rather than pre-emptive blocking. Software-based DRM prevents digital screenshots but not physical photography.

Screen-camera visible light communication (VLC) [14]–[16] similarly exploits human temporal integration to embed information imperceptible to the eye yet decodable by cameras. Our method is VLC's mirror image: VLC aims to let cameras *read* hidden information, whereas we aim to prevent cameras from *reading* displayed content.

Yerazunis and Carbone [17] first proposed temporal image masking for display privacy. Wu and Zhai [18] systematized temporal psychovisual modulation, and Gao et al. [19] applied it to information-security displays. Zhang et al. [5] proposed Kaleido, which re-encodes video frames via multi-frame decomposition to exploit the channel difference between display-eye and display-camera paths, making films "viewable but not recordable"; Yang et al. [20] extended this anti-piracy direction with Rainbow, covering mobile-camera filming in physical environments. These schemes degrade continuous recording quality, evaluating via PSNR, SSIM, and subjective quality scores on recorded footage, metrics not interchangeable with the OCR, detection, and VLM recovery rates used here. None address short-exposure single-frame capture of static sensitive content coupled with machine recognition. Mechanistically, Kaleido operates in continuous color space with chrominance-complementary frames and luminance variation, relying on perceptually complete mixing within a cycle; our method uses CSPRNG-driven discrete pixel-level partitioning, and the capture-hardened profile deliberately abandons strict completeness to resist integration attacks (§IV-E). Lim [21] used persistence of vision to generate anti-screenshot keyboards in a browser, where a single screenshot captures only partial patterns. Table 1 summarizes the key design-space differences between Kaleido-class temporal re-encoding and the present method.

TABLE 1. Theoretical Comparison of Temporal Display-Privacy Approaches

Dimension	Kaleido [5]	This work
Target threat	Continuous recording	Short-exp. single frame
Decomposition	Chrominance-compl.	Pixel-level binary mask
Cycle design	Perceptually complete	Hardened: breaks compl.
Full-cycle integ.	Recoverable by design	Deployed: recoverable; hardened: 47.9% char
Short-exp. single frame	Color info retained; single frame readable	5.0–15.1% char (OCR)
Evaluation target	PSNR/SSIM on video	OCR/VLM char recovery

Temporal partitioning per se is therefore not novel. Our incremental contributions are: (1) pixel-level CSPRNG complementary partitioning with a hardened profile targeting physical cameras and conventional OCR under short exposure; (2) 10,575 real captures covering 9 geometric conditions on a single UVC webcam, with the core short-exposure subset as feasibility evidence; (3) detection, tracking, and VLM probes as cross-task stress tests; and (4) quantitative boundary analysis of full-cycle averaging and VLM failure.

C. ADVERSARIAL PERTURBATIONS AND PHYSICAL-WORLD ROBUSTNESS

Digital adversarial perturbations [22]–[24] and physical-world adversarial patches [25]–[27] are surveyed by Wei et al. [28]. Athalye et al. [29] achieved the first cross-viewpoint-robust 3D physical adversarial objects via Expectation over Transformation (EoT), though their approach still requires white-box access to the target model. A large digital-to-physical robustness gap persists: pixel-level digital perturbations are weakened considerably after camera sampling and JPEG compression [25]. Physical adversarial patches exhibit cross-viewpoint robustness but require perturbation patterns pre-generated for specific classifiers [26], [27], with limited transferability to unseen models [30]. The screen-camera link also introduces moiré artifacts, which Niu et al. [31] demonstrated can themselves act as adversarial perturbations. Our method does not add perturbations to content; instead, it performs temporal partitioning at the display layer. Physical photography is inherently a sampling process, and the defense is embedded in the display itself, requiring no assumptions about the attacker's recognition model.

III. THREAT MODEL

A. PROTECTED ASSETS

Sensitive text and visual information displayed on screen: passwords, financial data, personally identifiable information (PII), source code, confidential documents, and similar content.

B. ATTACKER CAPABILITIES

Attack capabilities do not form a strict monotonic hierarchy but are jointly determined by exposure, geometry, temporal sampling, and post-processing capabilities. Results are reported across the following scope:

- **Primary protection scope**: Opportunistic short-exposure snapshots and per-frame recognition by conventional OCR attackers. Camera exposure time is shorter than or close to one subframe duration; the attacker may choose among OCR engines but does not possess observations covering a full display cycle. This tier represents a brief single-shot attempt without exposure calibration or sustained aiming, not the strongest rational attacker. Note that our experimental evaluation uses *fixed calibrated* exposures (3.91 ms) that are more favorable to the attacker than typical opportunistic auto-exposure on a bright display; the threat-model tier is defined by the single-frame constraint, not by the exposure-selection mechanism.
- **Evaluation boundary**: Long exposure, video temporal averaging after continuous recording, off-axis capture, phase search and channel reconstruction given a known period, object detection/association tasks, and end-to-end recognition by commercial vision-language models (§V-C). We quantify these risks but interpret capture-hardened results only as mitigation, not as a blocking guarantee.
- **Not yet adequately covered**: Additional camera/display combinations, precise rolling-shutter row alignment, real-world temporal super-resolution, learning-based reconstructors trained on display sequences, and long-term adaptive attacks.

C. ATTACKER KNOWLEDGE

The attacker *knows* the temporal masking algorithm, the subframe count n, and candidate period parameters, but does *not know* the current cycle's mask seed or phase starting point. The CSPRNG provides per-cycle mask randomization to prevent fixed-pattern learning and cross-cycle correlation; it does not provide a cryptographic guarantee against full-cycle averaging attacks. Once an attacker obtains and registers a complete cycle, an unknown seed cannot prevent linear reconstruction. More fundamentally, the method's security against short-exposure capture rests on disrupting the binarization and segmentation stages of conventional OCR pipelines, not on an information-theoretic impossibility of recovery from sparse pixels. End-to-end recognizers such as VLMs bypass binarization via visual encoding and language priors, and can recover substantial text from the same sparse fragments (§V-C).

D. PROTECTION GOALS

We define tiered design goals under short-exposure conditions by protection profile: the capture-hardened profile targets conventional OCR character recovery $< 5\%$ with exact-match $= 0\%$; the readability-priority deployed profile targets $\geq$80% relative reduction in character recovery with exact-match $< 1\%$. Changes in object detection mAP50 explore task-transfer risk beyond text and are not primary pass criteria. These thresholds guide parameter selection and profile trade-offs but are not formal security guarantees against all attack modes; VLM results are reported separately as security boundaries (§V-C) and do not enter pass/fail determination. On the usability side, FPI, ΔE, and SSIM are modeled proxies; because the planned user study has not been conducted, we make no "imperceptible to the human eye" claim.

IV. SYSTEM DESIGN

A. TEMPORAL SUBFRAME DECOMPOSITION

Given a normalized original frame $\mathbf{I} \in [0,1]^{H\times W\times C}$, we generate binary masks $\mathbf{M}_1, \ldots, \mathbf{M}_n$ satisfying $\sum_{k=1}^{n} \mathbf{M}_k = \mathbf{1}$, and set $\mathbf{I}_k = \mathbf{I} \odot \mathbf{M}_k$. This yields:

- **Digital-domain completeness**: $\sum_{k=1}^{n} \mathbf{I}_k = \mathbf{I}$;
- **Mutual exclusivity**: each pixel is illuminated in exactly one of the n time slots.

A distinction between "summation" and "temporal averaging" is necessary. On an ideal linear display with equal slot durations and equal peak luminance, the average radiance over one cycle is $\mathbf{I}/n$, not $\mathbf{I}$. Digital-domain completeness means subframes can be re-summed; it does not imply that real display luminance has been losslessly preserved.

Masks are generated using a ChaCha20 [32] CSPRNG and the Fisher–Yates shuffle [33], randomly assigning each pixel to one subframe. The implementation applies a chi-square uniformity check on generated slot counts (significance level $\alpha = 0.01$, df $= n-1 = 3$), re-sampling the seed upon failure (up to 5 retries). For $n = 4$, a power of two, each pixel's slot assignment draws from an unbiased 2-bit uniform distribution; the expected rejection rate equals the nominal $\alpha \approx 1\%$. Random seeds reduce fixed patterns and cross-cycle correlation but do not change the fact that a full cycle can be linearly summed. Fig. 2 illustrates the complete pipeline from original frame to protected display.

B. COMPLEMENTARY ADVERSARIAL NOISE

We overlay adversarial noise $\mathbf{N}_1, \ldots, \mathbf{N}_n$ on the subframes, satisfying:

$$\sum_{k=1}^{n} \mathbf{N}_k = \mathbf{0} \tag{1}$$

Equation (1) holds strictly only in the ideal linear domain prior to summation, clipping, and the display transfer function. Prior work has shown that small text-image perturbations can significantly mislead OCR systems [34], providing a basis for OCR-oriented noise design. Noise directions are generated by FGSM-style [22] proxy gradients with an amplitude bound of $\varepsilon = 8/255$, then decomposed complementarily in the temporal domain. Because displays cannot emit negative radiance, the implementation uses a pedestal level and numerical clipping. Consequently, after gamma correction, quantization, clipping, and camera imaging, the noise is not guaranteed to cancel physically.

Component role: The random dot-pattern mask is the primary mechanism. In controlled digital experiments, noise can suppress residual structure under weak masking or repair attacks. In real-capture short-exposure results, however, mask+noise (25.6%) actually exceeds mask-only (19.2%), and the Strong profile with weak stripe/glyph overlay also runs slightly higher (20.7% versus 19.2%) under the best-of-engine caliber. The noise is retained in the Strong/Deployed/Capture-hardened profiles despite this short-exposure negative result for two reasons: (1) it provides measurable benefit against integration attacks in the digital pipeline (temporal-average recovery drops from 95.2% to 89.7% in simulation when noise is present), and (2) it increases visual clutter in the captured frame, which empirically disrupts VLM glyph-completion priors on dense text (Table 7: CET6 recovery with noise-inclusive hardened profile is 0.4–18.5% versus unprotected 64.4%). The mask-only baseline (19.2% short-exposure) should be considered the cleaner reference for evaluating the temporal mask mechanism in isolation; noise is an optional layer whose net contribution is attack-mode-dependent.

C. LUMINANCE COMPENSATION

Each pixel is illuminated only once per n-slot cycle. If the display could boost light output by a factor of n in the linear-light domain during the active slot, the cycle-averaged luminance would approach the original. The current PoC implements neither per-slot dynamic backlight coordination nor photometric verification. All luminance alignment, CIEDE2000 ΔE_{00} [35], and SSIM results therefore come from digital integration/normalization models and serve only for cross-configuration comparison, not as evidence of real-panel optical equivalence or user comfort.

D. TIMING AND BANDWIDTH

For a basic cycle containing only n complementary subframes, taking 60 Hz as the minimum cycle rate requires $f_r \geq 60n$; at $n = 4$ this corresponds to 240 Hz. If the cycle additionally includes one inversion frame, the requirement becomes $f_r \geq 60(n+1)$, and at 240 Hz the actual cycle rate drops to only 48 Hz. Thus, 240 Hz meets the basic four-subframe configuration but not the inversion-frame configuration's 60 Hz cycle target. FPI is a modeled flicker proxy based on cycle rate and spatial pooling; lower values indicate lower predicted flicker risk rather than a clinical or user-study threshold. The spatial phase variation of random masks may reduce local synchronized modulation, but human flicker perception also depends on luminance, contrast, retinal eccentricity, and eye movements [36], [37]. Whether an observer integrates complementary subframes into a perceptually complete image depends on the temporal contrast sensitivity function (TCSF): at the 48 Hz full-cycle rate of the inversion configuration, the fundamental modulation frequency falls below the critical flicker fusion (CFF) threshold for high-contrast, foveal stimuli under typical photopic conditions (CFF ≈ 50–70 Hz [37]), meaning some observers may perceive flicker rather than a steady fused image, potentially violating the core premise of seamless temporal integration. The basic n=4 configuration without inversion (60 Hz cycle) sits near the CFF boundary and is less affected.

E. CAPTURE-HARDENED PROFILE

The basic linear temporal mask leaks under full-cycle temporal averaging (see §V-F2). To address this, we introduce nonlinear enhanced profiles. Unless otherwise stated, profiles in this subsection use $n = 4$ base temporal masking with complementary adversarial noise enabled. Table 2 summarizes all profile configurations; the "Mask + noise" row in Table 4 is the base condition without stripe/glyph anti-OCR overlay.

- **Strong (anti-OCR) profile**: base mask+noise with mask cell size of 1 pixel, stripe width of 10 pixels, stripe amplitude $\alpha_s = 0.10$, and glyph amplitude $\alpha_g = 0.12$, without an inversion frame.
- **Deployed profile**: the Strong profile plus an $\alpha = 0.2$ inversion frame, selected as the readability-priority configuration.
- **Capture-hardened profile**: base mask+noise with mask cell size increased to 2 pixels, stripe width of

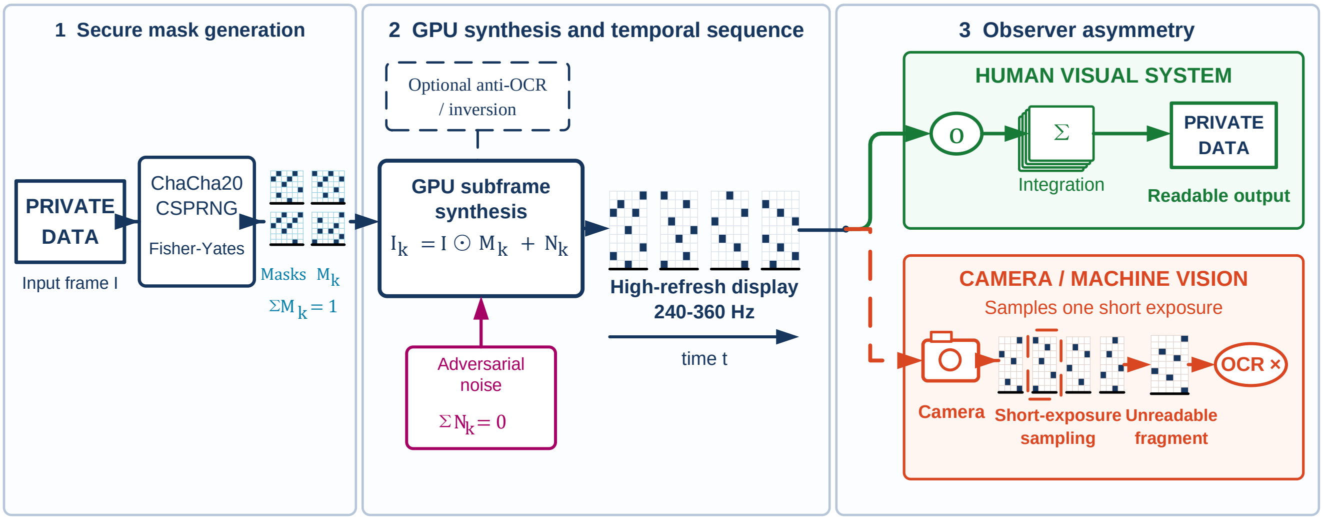

FIGURE 2. Temporal pixel masking system pipeline from source content to protected display. The input frame is decomposed into CSPRNG-assigned complementary pixel masks; optional complementary OCR-oriented noise, profile-dependent stripe/glyph overlays, and an optional partial inversion frame are then inserted before high-refresh playback. A human observer is expected to integrate multiple slots into a readable view, whereas a short-exposure camera may capture only a sparse subframe. Long exposure and video temporal averaging can accumulate multiple slots and are therefore evaluated separately as boundary attacks rather than assumed away.

TABLE 2. Profile Composition Summary ($n = 4$ for All Profiles)

Profile	Mask	Noise	Cell (px)	Stripe α_s	Glyph α_g	Inv. α
Mask only	✓	–	1	–	–	–
Mask + noise	✓	✓	1	–	–	–
Strong	✓	✓	1	0.10	0.12	–
Deployed	✓	✓	1	0.10	0.12	0.2
Capture-hardened	✓	✓	2	0.42	0.55	0.2

6 pixels, stripe and glyph amplitudes of 0.42 and 0.55 respectively, plus an $\alpha = 0.2$ inversion frame.

The nonlinear profiles depart from strict completeness ($\sum \mathbf{I}_k \neq \mathbf{I}$), producing visible stripes and grain. This trade-off is the opposite of Kaleido-class temporal re-encoding: the latter relies on perceptually complete mixing within a cycle to preserve viewing experience, meaning full-cycle integration inherently reconstructs an approximation of the original, consistent with its anti-piracy goal. In the static-content capture threat addressed here, completeness is precisely the entry point for integration attacks; the hardened profile therefore makes breaking strict completeness an active design trade-off, accepting visible stripes and grain in exchange for robustness to full-cycle integration. Its security benefit and unmeasured readability cost must be reported separately.

F. PARTIAL INVERSION FRAME

A long-exposure attack occurs when the exposure window covers a full display cycle ($n = 4$ at 240 Hz yields a basic cycle of approximately 16.7 ms; with an inversion frame the $n + 1$ slot cycle is approximately 20.8 ms; §V-A reports measured long-exposure times of 31.25 and 125 ms). Such attacks approximate the original image through sensor integration. A direct countermeasure inserts a **partial inversion frame** $\alpha \cdot (255 - \mathbf{I})$ after each n-subframe cycle, introducing a luminance component complementary to the original into the long-exposure integral and reducing net signal contrast. Higher α may reduce long-exposure recovery, but also alters the luminance and contrast of the digital integrated view.

A real-capture inversion strength ablation (Table 3; pipeline per §V-B) shows that weak inversion does not meaningfully reduce long-exposure recovery: $\alpha = 0$ yields 68.6% and $\alpha = 0.2$ yields 67.0% (best-of-engine), with 95% confidence intervals overlapping across $\alpha \in [0, 0.5]$. The ablation holds the Strong overlay fixed ($\alpha_s = 0.10$, $\alpha_g = 0.12$) and varies only inversion strength, using the five-item account/code/CET6 subset rather than the 12-item main corpus; its $\alpha = 0.2$ value ($N = 135$) is therefore not a repeated estimate of the deployed row's 60.9% ($N = 324$). Only near-full inversion ($\alpha = 1.0$) reduces recovery to 18.1%, at the cost of severely degrading the digital integrated view. Under video temporal averaging, weak inversion similarly maintains 64–73% recovery. Weak inversion alone cannot resist integration attacks. The deployed choice of $\alpha = 0.2$ is a modeled readability-priority design decision, not a user-study-validated optimum.

Each display cycle thus contains $n+1$ time slots. At $n = 4$ and 240 Hz the cycle rate is 48 Hz, below the 60 Hz design target and potentially producing perceptible flicker; no fixed "flicker-free threshold" applies universally across viewing conditions.

G. IMPLEMENTATION SCOPE

The current PoC operates at the application/algorithm layer, using Python and GPU rendering to generate display sequences. It does not include OS driver-level injection (DXGI / KMS-DRM / CoreDisplay) or per-slot dynamic backlight

TABLE 3. Inversion Frame Strength α Ablation on the Fixed Strong Overlay ($\alpha_s = 0.10$, $\alpha_g = 0.12$): Real-Capture Best-of-Engine Character Recovery (Mean and 95% Bootstrap CI, Five Account/Code/CET6 Items Pooled Across 9 Geometric Conditions; $N = 135$ per Setting for Long Exposure, $N = 45$ for Video Temporal Averaging). This Five-Item Subset Differs from the 12-Item Deployed Rows in Table 4.

α	Long-exp. char (%)	Video avg. char (%)
0.0	68.6 [63.1, 73.9]	72.7 [63.3, 81.1]
0.2	67.0 [61.3, 72.7]	69.8 [59.9, 78.7]
0.3	72.0 [66.2, 76.9]	64.3 [54.0, 73.6]
0.5	61.7 [55.6, 67.7]	66.4 [56.5, 75.2]
1.0	18.1 [14.0, 22.0]	28.4 [20.5, 36.9]

control. We therefore validate the display sequence and its captured results, without claiming production-level system integration.

V. EXPERIMENTAL EVALUATION

A. EXPERIMENTAL SETUP

Hardware platform: Experiments ran on a desktop workstation with an Intel Core i7-8700K (3.70 GHz), 32 GB RAM, and an NVIDIA TITAN Xp (12 GB VRAM). The display is an AOC 24G51Z (23.8-inch Fast IPS panel, QD backlight, edge-lit LED without local dimming, RGB stripe subpixels), driven at 1920×1080 with a nominal refresh rate of 240 Hz. The manufacturer specifies 1 ms gray-to-gray (GtG) response time; at 240 Hz each subframe slot lasts approximately 4.17 ms, so the GtG transition occupies roughly 24% of one slot. Slower gray-level transitions (particularly dark-to-light on IPS) may take 2–5 ms, causing partial temporal overlap (ghosting) between adjacent subframes. The measured real-capture results already include these panel-response effects; mechanistic narrative should therefore be read as "camera samples a short exposure that may span partial transitions across subframes" rather than "camera captures exactly one discrete subframe." This refresh rate satisfies only the basic $n = 4$ subframe cycle at $f_r/n = 60$ Hz. With an additional inversion frame per cycle, the actual cycle frequency is $240/(n{+}1) = 48$ Hz. Thus, the visual comfort of the inversion configuration cannot be inferred from "240 Hz high refresh rate" alone and still requires user-study validation.

Capture device: All physical captures use a single eMeet SmartCam S600 USB webcam (UVC standard driver). The playback canvas was 1920×1080; camera output underwent perspective correction and content cropping before recognition. Short-exposure and video-frame exposure times were 0.49 or 3.91 ms, long-exposure times 31.25 or 125 ms, and video was captured at 60 fps. The camera was placed at 0.5 m, 1 m, and 1.5 m from the display, rotated to 0°, 15°, and 30° at each distance, yielding 9 geometric configurations. Ambient illuminance was not recorded. The S600 uses a Sony 1/2.55-inch CMOS sensor with a rolling shutter. At 1080p/60 fps the frame period is 16.67 ms; the rolling-shutter readout time is bounded by this period and is typically 10–16 ms for consumer CMOS sensors at this resolution. Because the readout time far exceeds the 4.17 ms subframe duration ($n{=}4$ at 240 Hz), different image rows are exposed during different subframes within a single capture: the top and bottom of one "short-exposure" photograph may contain fragments from two or more subframes. This row-level subframe mixing is one mechanism explaining why real-capture short-exposure recovery (15.1%) exceeds simulated single-subframe recovery (0–2.6%), as it increases the total information content per frame relative to the ideal single-subframe assumption. Results apply to this UVC link and should not be extrapolated to smartphones, smart glasses, or other camera ISPs.

Test content: At each of the 9 geometric positions, the real-capture OCR experiment used a fixed set of 12 content items: 11 selected synthetic items and one CET6 (College English Test Band 6) document (portrait 900×1280). Three captures per item give 36 captures per profile–attack condition at each position and $12 \times 3 \times 9 = 324$ across all positions before the disclosed deployed-profile repeats. By contrast, the software-simulated OCR experiment used the complete 120-sample synthetic corpus (12 content templates $\times$ 10 variants, mapped to 9 category labels covering passwords, credentials, code, tables, paragraphs, URLs, etc.). Real-capture detection used a 150-image COCO val2017 [38] subset; tracking used 450 frames from the MOT17-09-FRCNN [39] sequence. Simulated detection used only 8 images for pipeline diagnostics; simulated tracking covered 5,316 frames. Real-capture short-exposure OCR is therefore the primary feasibility evidence; detection, tracking, and simulated detection results explore task-transfer risk, and the 8-image simulated detection is a pipeline diagnostic, not a full-scale evaluation.

Evaluation metrics: OCR character recovery is defined as $\max(0, 1 - \mathrm{CER})$. Exact-match is determined after whitespace removal and case-insensitive comparison of the full text. Sensitive-token recovery counts the proportion of reference-text digits, account numbers, URL/path strings, and key-like fields that are recovered in full. Leak rate is the proportion of samples with character recovery $\geq$20%, an arbitrary working threshold chosen as a round-number proxy for "non-trivial partial recovery"; no claim is made that 19% recovery is safe. To supplement this binary indicator, the per-capture recovery distribution is characterized by percentiles: for the deployed short-exposure condition, P50 = 8.3%, P75 = 16.7%, P90 = 43.0%, P95 = 60.9%, and P99 = 94.5% (N=459); for capture-hardened, P50 = 1.6%, P75 = 8.3%, P90 = 14.1%, P95 = 18.7%, and P99 = 22.2% (N=324). The heavy right tail of the deployed distribution indicates that a small fraction of captures (predominantly large-font short snippets at close range) yield near-complete recovery even under short exposure. Detection uses mAP, mAP50, and AR; tracking uses HOTA and IDF1, with MOTA/MOTP also reported in simulation. Section V-D defines the separate usability protocol and metrics.

For the real-capture OCR main results, we first take the maximum recovery across three engines per capture to model an attacker choosing the best recognizer, then estimate 95% bootstrap confidence intervals with 2,000 percentile resamples using captures as the resampling unit and a fixed random seed of 20260612. Confidence intervals for core conditions are reported inline in §V-B; ablation intervals appear in Table 3. Because repeated measurements exist for the same content and geometric conditions, these intervals describe empirical uncertainty within the current corpus and capture setup, not independent population inference. As a sensitivity check, a cluster bootstrap resampling by content item (12 clusters) yields wider intervals for the deployed short-exposure mean: [11.5, 18.6]% versus [13.3, 17.1]% with capture-level resampling. Excluding the 0.49 ms underexposed position (d0.5/15°) raises the deployed short-exposure estimate to 16.7% [14.7, 18.7]%, a conservative shift that does not change qualitative conclusions. A leave-one-round-out analysis on the five repeated content items shows stability: round 1 yields 16.6%, round 2 yields 15.8%, both within the cluster CI. Detection and tracking results have not undergone significance testing; differences are reported descriptively.

Digital usability proxies: ΔE_{00} is the mean CIEDE2000 color difference between the original image and the full-cycle digitally reconstructed image; SSIM is their mean structural similarity. The normalized mutual information proxy is defined as $I(X;Y)/H(X)$, where X is the original grayscale and Y is a single-subframe grayscale, estimated with 64 histogram bins; lower values indicate less information retained by a single subframe. FPI uses a simplified model in our implementation: let single-pixel modulation depth $d = 1 - 1/n$, spatial pooling pixel count $N = 625$, and pixel illumination frequency $f_p = f_r/n$. When $f_p \geq 60\,\text{Hz}$, FPI $= dN^{-1/2}(60/f_p)^2$; below 60 Hz the formula uses $d(1-f_p/60)$. Note that this piecewise form is discontinuous at $f_p = 60\,\text{Hz}$: in a narrow band just below 60 Hz the low-frequency branch yields values smaller than the high-frequency branch at 60 Hz, contradicting the intuition that lower frequency means worse flicker. The 12 configurations swept here (f_p grid points from 18–180 Hz) do not land in this narrow band, so configuration ranking is unaffected. This formula is not derived from IEEE Std 1789-2015 [40] and does not constitute a clinical or perceptual safety threshold; it serves only for within-paper configuration ranking.

B. REAL-CAPTURE OCR EXPERIMENT

The real-capture OCR corpus comprises 10,575 images (1,175 per geometric position), all from the S600 webcam. The corpus mixes component ablations, parameter sweeps, and main attack conditions; it is not a balanced "9 geometries × 6 main profiles × 3 attacks" full-factorial design. One asymmetry requires disclosure: the deployed profile performed two capture rounds for 5 of the 12 content items at each position (2 account credentials, 2 code snippets, 1 CET6 document), giving 459/153 short-exposure/video samples rather than 324/108; both rounds enter analysis. Three OCR engines, Surya [41], EasyOCR [42], and Tesseract [43], were applied, producing 31,725 engine-level records. Recent VLMs show strong OCR capabilities [44]; the probes in §V-C confirm that protection against conventional OCR does not transfer to VLM attackers. Table 4 reports core conditions with per-capture best-of-engine; Table 5 gives per-engine short-exposure results. Fig. 3 shows qualitative comparisons; Fig. 4 summarizes recovery rates by profile and attack mode.

Key findings:

1) **Short-exposure recovery drops substantially**: Unprotected screen photographs yield 94.1% OCR character recovery (95% CI [93.1, 95.0]; exact 65.7%). The Strong profile achieves 20.7% char / 0.6% exact; the deployed profile (Strong + $\alpha = 0.2$ inversion) reaches 15.1% (CI [13.3, 17.1]; exact 0.4%), an 83.9% relative reduction meeting the goal in §III. The 5.6-pp gap between Strong and deployed is descriptive only: the deployed profile has $N = 459$ (with repeated captures) versus 324, so we do not interpret it as a strict causal effect. The capture-hardened profile reduces recovery to 5.0% (CI [4.3, 5.8]; exact 0%); exact-match meets the strict pass criterion, but the character recovery target of $<5\%$ is *not conclusively met*: the point estimate equals the threshold and the 95% CI upper bound (5.8%) exceeds it. The result is consistent with the target being approximately reached but cannot be statistically distinguished from marginal exceedance. These results support short-exposure mitigation feasibility within the tested camera, exposure, and geometry range.
2) **Integration attacks remain the residual frontier; long-exposure reversal is a notable negative result**: Both long exposure and video temporal averaging exploit full-cycle complementarity. The deployed profile still recovers 60.9% char / 36.4% exact under long exposure (sensitive-token recall 96.5%) and 71.1% char / 42.5% exact under video averaging. The divergence between character and sensitive-token recovery reflects the structural properties of high-value fields: digits, account numbers, and URLs are typically short, rendered in large font with a restricted character set, and thus recoverable from degraded images even when surrounding dense text is not. **Long-exposure reversal**: the unprotected long-exposure baseline is only 47.3% char, lower than the deployed profile's 60.9%. Temporal masking *assists* the attacker under long exposure by reducing effective luminance into the sensor's dynamic range. Splitting by exposure time reveals the mechanism: the sole 125 ms position (0.5 m/15°) yields only 1.1% unprotected recovery (sensor saturation on the full-brightness screen) versus 46.3% deployed (temporal masking reduces effective luminance to $\sim 1/n$, keeping the image within the sensor's dynamic range). Excluding this position, at 31.25 ms the anomaly persists at the pooled level (unprotected 53.0% versus deployed 62.7%) but is position-dependent: at 0.5 m the expected direction holds (unprotected 78–82% $>$ deployed 25–55%), whereas at 1.0–1.5 m several positions show deployed recovery exceeding unprotected by 12–59 pp. We attribute this to the stripe/glyph overlays providing

TABLE 4. Single-UVC Real-Capture OCR Core Conditions (Best-of-Engine; N = Independent Captures per Row; Pooled Across 9 Geometric Conditions)

Profile	Attack mode	N	Char recovery	Exact match	Sensitive token	Leak rate
Original (unprotected)	short	324	94.1%	65.7%	98.4%	99.4%
	video:temporal_mean	108	79.9%	61.1%	85.6%	86.1%
	long	324	47.3%	18.8%	95.7%	58.6%
Mask only	short	324	19.2%	0.3%	30.8%	30.6%
	video:temporal_mean	108	79.0%	48.1%	84.0%	88.0%
	long	324	67.2%	35.5%	94.0%	75.0%
Mask + noise	short	324	25.6%	2.8%	34.4%	36.1%
	video:temporal_mean	108	79.2%	49.1%	83.9%	88.9%
	long	324	68.0%	39.8%	95.7%	76.2%
Strong (anti-OCR)	short	324	20.7%	0.6%	33.7%	32.4%
	video:temporal_mean	108	78.6%	50.9%	83.3%	88.9%
	long	324	61.6%	39.8%	95.7%	71.3%
Deployed	short	459	15.1%	0.4%	24.0%	20.5%
	video:temporal_mean	153	71.1%	42.5%	78.5%	85.0%
	long	324	60.9%	36.4%	96.5%	69.1%
Capture-hardened	short	324	5.0%	0.0%	6.5%	3.7%
	video:temporal_mean	108	47.9%	5.6%	45.6%	76.9%
	long	324	9.3%	0.0%	34.9%	14.2%

residual edge structure that assists OCR at geometries where the unprotected image suffers from local overexposure or contrast compression with the fixed 31.25 ms setting. This anomaly does not weaken short-exposure conclusions (where deployed is always lower) but demonstrates that fixed long-exposure parameters produce nonmonotonic effects across display conditions. Weak inversion ($\alpha = 0.2$) alone is insufficient to resist temporal integration. The weak Strong overlay likewise provides no measurable integration defense: mask-only and Strong are nearly identical under video averaging (79.0% versus 78.6%) and differ only modestly under long exposure (67.2% versus 61.6%), with overlapping bootstrap intervals. The monotone synthetic trend in Fig. 5 therefore does not imply a real-capture integration gain at weak overlay amplitude; measurable reduction appears only at capture-hardened amplitudes, and even there leakage remains substantial. The primary validity conclusions are thus confined to single-frame short exposure; integration attacks are reported as a security boundary.

3) **Capture-hardened profile: reduced integration leakage with unresolved perception cost**: This profile yields 9.3% char (CI [7.9, 10.9]) / 0% exact for long exposure and 47.9% char (CI [41.8, 54.0]) / 5.6% exact (CI [1.9, 10.2]) for video averaging. The video leakage is substantial and does not constitute attack closure. The pooled 47.9% masks large inter-position variation: 0.5 m at three angles yields 50.7%/0%/33.8%; 1.0 m yields 61.1–69.4%; 1.5 m yields 48.4–55.5%. The 0% position (0.5 m/15°) was calibrated with the shorter 0.49 ms video-frame exposure (§V-A), producing visibly underexposed frames; all other positions use 3.91 ms, with recovery ranging nonmonotonically from 33.8–69.4%. Deployment assessment should therefore use worst-case geometry and exposure. SSIM from the digital pipeline reveals a proxy trade-off (Fig. 5) but is not real-capture or human-subject evidence. The user study protocol (§V-D) excludes this profile, leaving its perceptual cost unquantified.

4) **Geometric trends**: Across 9 conditions (0.5–1.5 m × 0–30°), protected short-exposure recovery is consistently below unprotected values. Because the design is unbalanced and no inferential model for geometric factors has been fitted, we do not claim statistically significant monotonic robustness in distance or angle.

Per-engine results: Table 5 breaks down short-exposure results by engine; Fig. 6 visualizes all six profiles. The three engines differ widely in unprotected recovery (Surya 37.1%, EasyOCR 79.5%, Tesseract 84.3%). Under Strong, recovery ranges from 2.1–16.3%; under deployed, 3.2–11.7%; under capture-hardened, 1.8–2.0%. All three engines show reduction, but with only three engine families the finding is not recognizer-agnostic. Best-of-engine takes the maximum across engines per capture, modeling an attacker who picks the best recognizer; it is the more conservative caliber from the defender's perspective. Because the maximum biases upward with each additional engine, the three-engine ceiling likely underestimates a real attacker with access to more engines or optimized pipelines; the per-engine average in Table 5 provides a less biased but less conservative reference.

C. REAL-CAPTURE VLM PROBES

To test whether conventional OCR conclusions extend to stronger recognizers, we applied three commercial VLMs (Qwen3-VL-32B-Instruct, Kimi-K2.6, and GLM-4.5V, via SiliconFlow API) to the same real-capture photographs. This section is a *boundary probe*: if VLMs recover text where conventional OCR fails, the practical scope of the method is confined to conventional OCR short-exposure mitigation.

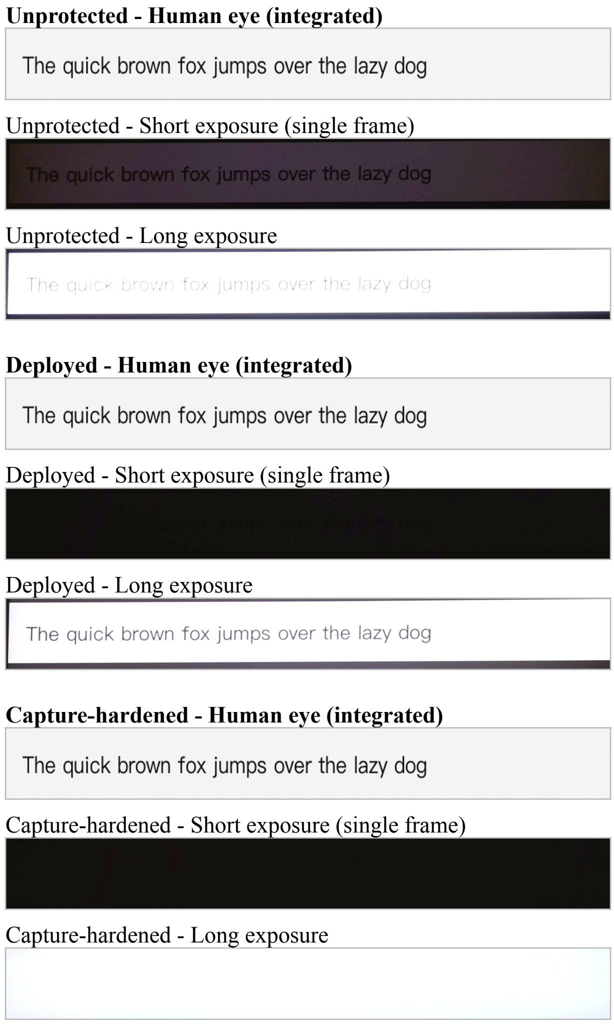

FIGURE 3. **Real-capture montage. Within each vertically stacked profile group (unprotected, deployed, and capture-hardened), panels show, from top to bottom, the digital integrated reconstruction view (for checking full-cycle image consistency), short-exposure capture, and long-exposure capture. Digital integrated views are not direct evidence of human readability or visual comfort.**

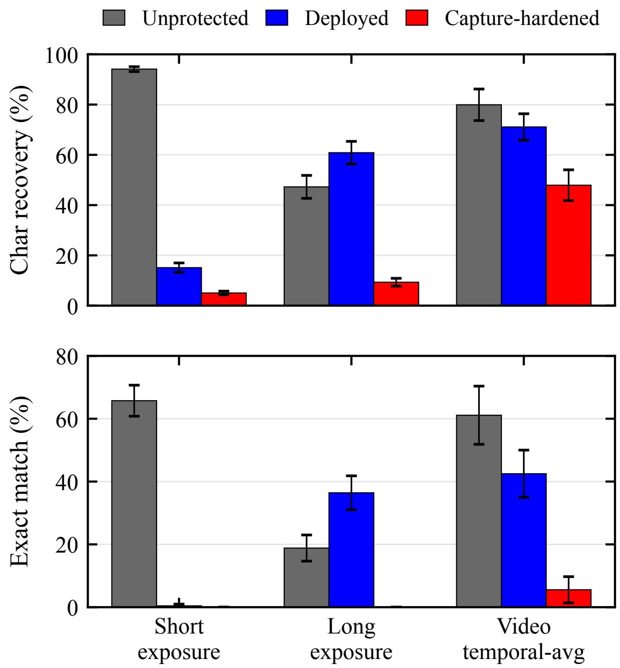

FIGURE 4. **Best-of-engine OCR character recovery and exact match for three representative real-capture profiles (unprotected, deployed, and capture-hardened) under short exposure, long exposure, and video temporal averaging, pooled across the 9 tested distance–angle configurations. Lower bars indicate less recovered text. The long-exposure and video bars expose the integration boundary: the readability-priority deployed profile retains substantial recoverable content even when its short-exposure bar is low. Deployed sample counts include the disclosed five-item repeat and therefore differ from the other profiles.**

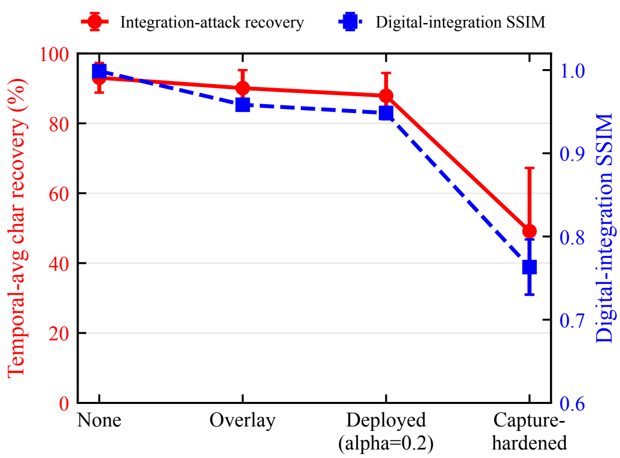

FIGURE 5. **Trade-off between digital reconstruction quality and integration-attack robustness: as anti-OCR enhancement increases (no overlay $\rightarrow$ strong overlay $\rightarrow$ deployed (overlay + $\alpha = 0.2$ inversion) $\rightarrow$ capture-hardened), temporal-average character recovery decreases (left axis) and full-cycle digital reconstruction SSIM [45] also decreases (right axis). Data from anti-OCR profile ablation on synthetic corpus (digital pipeline, not real capture); SSIM is an image proxy, not a user readability measure.**

The probes cover 0° on-axis at three distances. The 0.5 m run includes 156 inputs: 36 unprotected short-exposure baselines, 36 capture-hardened short/long-exposure each, and 4 types of aggregate views from 60 fps video (12 segments each). The 1.0 m and 1.5 m runs each include 120 inputs for the capture-hardened profile: 36 short/long-exposure each plus 4 video aggregate types (12 segments each). VLM inputs and the OCR best-of-engine (BoE) reference column in Table 6 use the same file batch (short/video-frame exposure 3.91 ms, long exposure 31.25 ms). All inputs are raw camera JPEGs without contrast enhancement; pilot tests showed that CLAHE-type local enhancement partially reconstructs masked text, turning preprocessing into an attack component. Of 1,188 model requests, 1,133 returned parseable results: 34 failures at 0.5 m (468 calls; Qwen 2, Kimi 0, GLM 32), 21 at 1.0 m (360 calls; Qwen 4, Kimi 1, GLM 16), and zero at 1.5 m (360 calls). Each cell reports effective N. GLM failures concentrate on mixed-content (47–54%), digit-string (23–38%),

TABLE 5. Real-Capture Short-Exposure OCR: Per-Engine Results (Pooled Across 9 Geometric Conditions)

Profile	Engine	Char (%)	Exact (%)	N
Original	Surya	37.1	22.8	324
	EasyOCR	79.5	27.8	324
	Tesseract	84.3	47.8	324
Mask only	Surya	4.2	0.3	324
	EasyOCR	15.6	0.0	324
	Tesseract	3.2	0.0	324
Mask + noise	Surya	11.0	2.8	324
	EasyOCR	18.1	0.0	324
	Tesseract	4.8	0.0	324
Strong	Surya	6.1	0.6	324
	EasyOCR	16.3	0.0	324
	Tesseract	2.1	0.0	324
Deployed	Surya	3.2	0.4	459
	EasyOCR	11.7	0.0	459
	Tesseract	3.3	0.0	459
Capture-hardened	Surya	1.8	0.0	324
	EasyOCR	2.0	0.0	324
	Tesseract	1.9	0.0	324

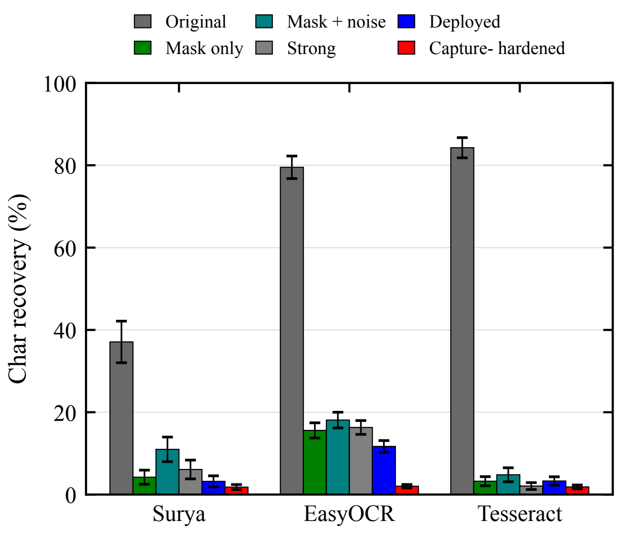

FIGURE 6. Real-capture short-exposure character recovery for three OCR engines across all six profiles in Table 5; error bars are 95% bootstrap intervals resampled by capture. All three engines are suppressed, but with a limited engine family this cannot be claimed as recognizer-agnostic.

and account-credential (31%) items, precisely the short high-value snippet types, suggesting failures correlate with content difficulty rather than occurring randomly. For the GLM 0.5 m short-exposure cell ($N_{\mathrm{eff}} = 31$), worst-case imputation (failures $=$ 0%) gives 61.0% char / 44.4% exact; best-case (failures $=$ 100%) gives 74.9% / 58.3%, versus the reported 70.8% / 51.6%. We use the failure-free 1.5 m run as primary evidence; the 1.0 m run enters Table 6's short-exposure distance sweep as supplementary data only. At 1.0 m, Qwen and Kimi long-exposure exact are 84.4% and 74.3%, and all three models' video temporal-average exact ranges from 70.0–83.3%, consistent with 1.5 m trends. GLM long-exposure exact at 1.0 m is only 3.3% (versus 58.3% at 1.5 m), so we make no cross-distance consistency generalization; per-record results are in the data archive. For compactness, Table 6 reports only the temporal-mean video aggregate at 0.5 m; the 1.5 m run reports all four aggregate views to show view-selection sensitivity. Model identifiers are Qwen/Qwen3-VL-32B-Instruct, Pro/moonshotai/Kimi-K2.6, and zai-org/GLM-4.5V; decoding temperature 0, maximum output 4,096 tokens. All three share a transcription prompt: transcribe only visible text, do not guess missing characters; ground truth is not provided. Full prompts, parameters, and per-call outputs are archived (see "Data and Code Availability"). Metrics are computed per §V-A, pooled at the capture level. CET6 documents are publicly available and may inflate recovery through training-data memorization; synthetic snippets and exact-match are therefore the cleaner discriminative caliber.

0.5 m session selection transparency: The archive preserves an alternative session (`real_capture_vlm_d0.5_a0_rerun.json`, timestamp 012715) at the same position whose capture-hardened short-exposure inputs are nearly all-black, yielding empty transcriptions and 0% char/exact across all three models. This session is archived as an under-exposure diagnostic but does not enter Table 6. We report the 141107 (3.91 ms) session because it shares the OCR reference batch and has adequate exposure, i.e., the condition more advantageous to the attacker. These two sessions are the only 0.5 m on-axis VLM capture sessions conducted; no other candidates exist. The selection follows input alignment and conservative reporting, not outcome-based cherry-picking.

Table 6 presents the pooled VLM probe results: short-exposure rows show distance variation, while long-exposure and video aggregate rows show integration attack boundaries. Table 7 breaks down 0.5 m capture-hardened short-exposure character recovery by content type.

Key findings:

1) **VLMs bound rather than extend the claim**: On the same 0.5 m photographs, best-of-engine OCR yields only 8.4% char / 0% exact on the hardened short-exposure condition, while Qwen3-VL achieves 91.5% char / 77.8% exact. At 1.5 m, the three models retain 33.3–47.2% exact / 61.5–70.0% char. Even with a prompt prohibiting guessing, VLMs recover substantial text from visible fragments. The conclusion is not that the hardened profile works against stronger attackers, but that conventional OCR claims do not transfer to VLM attackers.
2) **Failure concentrates on large-font short snippets**: At 0.5 m, CET6 full-page dense documents yield 0.4–18.5% VLM character recovery (unprotected baseline 64.4%), whereas credentials, code, and digit strings yield 50–100% (Table 7). At 1.5 m, CET6 dense pages drop to 0% char across all three models, while digit strings and short sentences remain highly recoverable. The pattern matches mask granularity: small-font

TABLE 6. Real-Capture VLM Probes (0° On-Axis; All Rows Except the Original Short Baseline Are Capture-Hardened). Each VLM Cell Reports Exact / Char / *N* (% / % / Effective Calls); OCR BoE Column Shows Same-Batch Three-Engine Best-of-Engine Reference (Exact / Char, %). For compactness, 0.5 m reports only the temporal-mean video aggregate; the remaining 0.5 m aggregate views are archived, while 1.5 m reports all four aggregate views.

Distance	Condition	OCR BoE	Qwen3-VL-32B	Kimi-K2.6	GLM-4.5V
Short-exposure distance sweep					
0.5 m	original short	58.3 / 92.4	83.3 / 94.5 / 36	83.3 / 95.6 / 36	73.1 / 92.0 / 26
0.5 m	short	0.0 / 8.4	77.8 / 91.5 / 36	47.2 / 84.7 / 36	51.6 / 70.8 / 31
1.0 m	short	0.0 / 3.6	55.6 / 79.1 / 36	38.9 / 68.6 / 36	27.3 / 47.0 / 33
1.5 m	short	0.0 / 6.5	47.2 / 68.9 / 36	33.3 / 61.5 / 36	36.1 / 70.0 / 36
Key attack views					
0.5 m	long	0.0 / 2.1	82.4 / 85.7 / 34	2.8 / 7.9 / 36	0.0 / 4.9 / 30
0.5 m	video:temporal_mean	8.3 / 50.7	83.3 / 94.0 / 12	83.3 / 95.5 / 12	87.5 / 91.8 / 8
1.5 m	long	0.0 / 13.2	91.7 / 89.3 / 36	88.9 / 89.9 / 36	58.3 / 62.4 / 36
1.5 m	video:single_best	0.0 / 4.3	8.3 / 24.8 / 12	0.0 / 12.7 / 12	0.0 / 29.8 / 12
1.5 m	video:temporal_mean	0.0 / 48.4	66.7 / 90.3 / 12	66.7 / 90.3 / 12	58.3 / 88.5 / 12
1.5 m	video:window_mean_best	0.0 / 8.6	58.3 / 89.0 / 12	41.7 / 84.2 / 12	66.7 / 87.2 / 12
1.5 m	video:max_proj	0.0 / 19.0	66.7 / 90.1 / 12	58.3 / 89.2 / 12	58.3 / 88.7 / 12

TABLE 7. 0.5 m Capture-Hardened Short-Exposure VLM Character Recovery by Content Type. Baseline Column Shows Same-Session Unprotected Original Short Results for Qwen (Kimi/GLM Similar); GLM Had Occasional Call Failures, Parenthesized *N* Shows Effective Count. At 1.5 m, CET6 Dense Pages Yield 0% Char Across All Three Models.

Content type (N)	Baseline (%)	Qwen (%)	Kimi (%)	GLM (%)
CET6 document (3)	64.4	18.5	5.6	0.4
Credentials (6)	97.6	97.6	88.3	77.1 (5)
Code snippets (6)	89.8	94.9	95.8	88.9
Digit strings (6)	100.0	100.0	98.6	50.0 (4)
Mixed content (6)	100.0	100.0	92.5	98.2 (4)
Paragraphs (6)	100.0	99.8	81.7	66.5
English sentences (3)	95.3	95.3	96.9	94.6

strokes (∼1–2 pixels wide) lose glyph topology when $1-1/n$ of pixels are removed. Large-font glyphs retain global contours even through sparse fragments, and VLMs can complete them via visual encoding and language priors [44], [46], bypassing the binarization bottleneck that disrupts conventional OCR.

3) **Attack capability varies across models and conditions**: Under 0.5 m long exposure, Qwen3-VL recovers 82.4% exact while Kimi/GLM do not exceed 2.8%; at 1.5 m, long-exposure exact rises to 58.3–91.7%. VLM screen-reading ability depends jointly on model version, distance, exposure, and content; any conclusion about "defeating current VLMs" will expire as models update.

4) **Video temporal averaging is the VLM residual frontier**: At 0.5 m, temporal-average yields 83.3–87.5% exact / ≥91.8% char; the three other 0.5 m aggregate views (single-best, window-mean-best, and max-projection) yield Qwen exact of 66.7%, 83.3%, and 83.3% respectively, confirming that view selection matters but all multi-frame aggregates far exceed single-frame performance. At 1.5 m, 58.3–66.7% exact / 88.5–90.3% char. By contrast, 1.5 m single-best-frame yields only 0–8.3% exact, and conventional OCR on the same temporal-average views stays at or below 8.3% exact (48.4–50.7% char). For VLM attackers, full-cycle integration (§V-F2) is far more dangerous than selecting a single clear frame, and does not require precise alignment or complex de-striping.

The effectiveness claim must therefore be bounded by content and attacker type: the hardened profile protects dense small text against conventional OCR but fails for large-font short snippets (credentials, code, digit strings) when the attacker uses a modern VLM. This negative result repositions the practical scope from "blocking arbitrary machine screen-reading" to "reducing conventional OCR short-exposure batch recovery," and identifies VLM-aware display protection as an open problem. Countermeasure directions are discussed in §VI-E. The section covers a single camera at 0° on-axis, one profile, three commercial API models, and only 12 video segments per cell; numbers are descriptive boundary probes, not full cross-geometry or cross-model-version evaluation.

D. USER EXPERIENCE STUDY

To evaluate the impact of the protected display on human readability and visual comfort, we designed a web-based user study.

Equipment and environment protocol: The same 240 Hz display as in §V-A. The formal study version sets a hard measured refresh-rate threshold of ≥200 Hz to ensure the fixed $n = 4$ basic subframe cycle rate $f_r/n \geq 50$ Hz; below this threshold participation is blocked rather than silently reducing n. The 144–199 Hz adaptive path is available only for an explicitly labeled demonstration mode, and its sessions are excluded from analysis by default. Before each participant begins, the experimenter confirms fixed display brightness, disabled auto-brightness/power-saving and variable refresh rate, unified browser fullscreen, closed high-load background processes, and an approximate 60 cm viewing distance. Refresh rate is re-measured after any display-mode change.

The browser records actual frame intervals in each trial's `requestAnimationFrame` loop; if the interval between adjacent callbacks exceeds $1.5\times$ the expected interval computed from the pre-trial measured refresh rate, a dropped-frame event is logged, along with estimated drop count, drop rate, mean/max frame intervals, observed refresh rate, basic subframe cycle rate, and full cycle rate with inversion frame. This recording reflects only browser presentation cadence and cannot substitute for panel scanning or photometric measurement.

Ethics and safety: The planned study involves minimal-risk human-subjects research. The authors' institution does not maintain an ethics review board for non-medical behavioral experiments of this type; accordingly, the protocol specifies informed consent and the safety measures described below, but no formal data collection has yet been conducted. Before starting, participants must separately check informed consent and photosensitivity screening: the page explicitly states that the masked display involves rapid temporal modulation, that participants should stop immediately upon experiencing discomfort, eye fatigue, dizziness, nausea, or headache, and that they may withdraw unconditionally at any time. Individuals who self-report a history of photosensitive epilepsy or sensitivity to flicker are not permitted to proceed. The system records consent timestamps and screening pass flags. Implementation-level safety is ensured by a $f_r/n \geq$ 50 Hz floor; at $n = 4$ on a 240 Hz panel the basic cycle rate is 60 Hz. The inversion frame reduces the full cycle frequency to $f_r/(n+1) = 48$ Hz; the implementation separately logs warnings and actual timing. Registration information (student ID, name) is used only for participation management; analysis and publication use only de-identified data. Age and gender are optional demographic fields.

Task design: Each participant first completes one 12-second, unscored unmasked Canvas typing warm-up, views a 10-second, unscored deployed full-mask preview, and then completes an 8-second, unscored typing practice under the same deployed mask. This sequence separates initial stimulus and input adaptation from the four subsequent 20-second scored trials (within-subjects design). On every typing screen, including warm-up, practice, and scored trials, the stimulus Canvas is visible and its player has started before the participant clicks "Start." Clicking "Start" triggers a 5-second countdown while the input remains disabled; the timed typing interval and input availability begin only when the countdown ends. The same pre-start viewing procedure applies to both control and masked conditions. Two conditions are each repeated twice, counterbalanced using ABBA or BAAB order based on the experimenter's continuously assigned zero-based sequence number k to control for practice and fatigue effects: (A) **original text** (control), displayed as unmasked Canvas with $n = 1$; (B) **masked text**, using the actual deployment configuration: $n = 4$, mask+noise, strong anti-OCR ($\alpha_s = 0.10$, $\alpha_g = 0.12$, 6 cycles) + weak inversion ($\alpha = 0.2$). Specifically, the typing order index is $k \bmod 2$ and the subjective rating Latin-square row is $\lfloor k/2 \rfloor \bmod 6$; every consecutive 12 participants thus fully cover the 2×6 joint assignment, which produces deterministic and balanced joint order allocation. Formal sessions enforce a uniqueness constraint on k in the database, and the backend recomputes and verifies both order indices. Both conditions are rendered by the same `renderSourceCanvas` path producing light text on a dark background, with identical canvas size, 23 px bold font, and color; the sole processing difference is the temporal protection pipeline, avoiding polarity and font-weight confounds. Here "cycles" refers to the web implementation's rotation of mask, noise, and stripe patterns through distinct instances before repeating; a value of 6 prevents long-exposure photography from integrating a single fixed pattern into a recognizable residue, sharing α_s and α_g parameters with the real-capture configuration described in §IV but constituting a web-playback-specific setting.

Scoring no longer compares characters position-by-position but aligns the full transcription against the optimal target prefix using Levenshtein minimum string distance (MSD) [47]; untyped target suffix does not count as error. From the alignment, accuracy, CPM, and WPM are computed, along with edit distance, aligned prefix length, and scoring version. First-keystroke latency is defined from the end of the 5-second countdown, when the input is enabled and the scored timer starts, to the first input event that makes the transcription non-empty. Analysis first averages each participant's two trials per condition, then performs within-subject paired comparison.

Subjective ratings: After the typing trials, participants rate 6 conditions on the four 1–5 Likert items displayed by the interface: readability (1 = difficult to read, 5 = very clear), stability (stored in the `flicker` field; 1 = strong flicker, 5 = no perceptible flicker), immediate visual comfort (`fatigue`; 1 = very uncomfortable, 5 = very comfortable), and subjective anti-peeking effectiveness (`privacy`; 1 = a bystander can read the display very easily, 5 = a bystander cannot read it at all). Higher values therefore indicate a better outcome on every item, including greater stability and comfort. Conditions include the unmasked original anchor, $n \in \{2, 3, 4\}$ mask+noise, $n = 4$ mask-only, and the same $n = 4$ deployed full configuration (mask+noise+strong anti-OCR+weak inversion) used in the typing task. The first three masking levels all achieve $\geq$60 Hz real temporal playback on a 240 Hz panel; the originally planned $n = 8$ would yield 30 Hz and trigger static fallback, so it is not included as a temporal ablation. The six conditions are presented in a 6th-order balanced Latin square; each condition must be viewed for at least 10 seconds before submission is enabled, with viewing start/end times and duration recorded. The unmasked anchor is also an attention/scale-interpretation aid but is not mechanically excluded by a "must rate 5" rule. The anti-peeking item is the participant's subjective estimate of bystander legibility, not an objective shoulder-surfing test with independent observers or controlled viewing geometry. This study does not include the capture-hardened profile; assessment of its visible stripes and grain's acceptability is left to future work (see §VI).

Sample size and analysis plan: A minimum of $N = 24$

participants is planned, a number within the range required for paired effect sizes of $d_z \approx 0.6$–0.8 at 80% power, and exactly covering two rounds of the 12-person joint order assignment. Pre-registered exclusion criteria are debug/demo sessions, incomplete four typing or six rating records, measured refresh rate below 200 Hz, any typing trial with fewer than 5 attempted characters, control mean accuracy below 50%, masked trials not in temporal mode, any trial with observed basic cycle rate below 50 Hz, or minimum-view straight-line ratings. The last criterion is defined as all six rating rows having a recorded viewing duration no greater than 11 seconds (a 1-second tolerance above the 10-second submission gate) and, within every row, identical scores on all four items. WPM, CPM, accuracy, attempted characters, and first-keystroke latency are first averaged within each participant by condition; speed metrics must be interpreted jointly with accuracy to avoid misinterpreting fast input under low accuracy as performance improvement. Paired differences are tested using paired t-tests when the pre-specified Shapiro test is passed, otherwise Wilcoxon signed-rank tests, with effect sizes and participant-level bootstrap 95% confidence intervals reported. Likert data are analyzed per dimension using Friedman tests with Holm-corrected post-hoc pairwise Wilcoxon tests. Analysis scripts generate de-identified participant means, JSON audit reports, and two LaTeX tables directly from the formal database `study_formal.db`.

Participants: **[TODO: Final participant count and demographics.]**

[TODO: Table: Typing trial objective metrics (accuracy/CPM/WPM, control vs. masked, paired differences + CI).]

[TODO: Table: 6 conditions × 4 dimensions Likert means + SD.]

[TODO: Core finding: whether the deployed full configuration produces significant changes in typing accuracy/speed and first-keystroke latency; subjective readability, stability, immediate visual comfort, and anti-peeking impression ratings.]

E. REAL-CAPTURE OBJECT DETECTION AND TRACKING

To probe whether the temporal degradation transfers beyond text, we report an exploratory detection/tracking stress test within the same camera link; evidence strength is not equivalent to the OCR main experiment. Content was displayed on the 240 Hz screen, captured by the S600 at 1.5 m / 0°, and processed by four detectors [48]–[51]. Detection uses real_clean (unprotected capture through the same link) as reference (Table 8). Tracking uses 450 frames from MOT17-09-FRCNN via still-display-then-capture rather than continuous video synchronization (Table 9); the associator is a greedy-fallback implementation inspired by ByteTrack [52], not official ByteTrack. The real_clean reference partially controls for link degradation, but exposure interactions with the protected display may persist.

Under short exposure, all four detectors' mAP50 drops from 39.2–50.2% to 1.6–5.8% (Fig. 7), relative decreases exceeding 85%. The real_video condition yields 3.0–7.3%, also below the real_clean baselines, consistent with visual feature degradation beyond character-segmentation-based OCR. With only one camera, one position, and 150 images, these results provide directional evidence rather than a cross-device detection-protection conclusion. In still-frame tracking, HOTA drops from 14.2–15.8% to 5.1–8.7% and IDF1 from 9.6–12.1% to 4.4–6.5%. The real_clean baseline HOTA (14.2–15.8%) is itself depressed by the still-frame link, far below native-image levels, compressing the room for further decrease. The still-frame procedure cannot substitute for continuous-video tracking.

TABLE 8. Real-Capture COCO Detection (150 Images; real_clean = Unprotected Capture Baseline)

Detector	Condition	mAP	mAP50	AR
YOLO26x	real_clean	28.6%	43.9%	31.0%
	real_short	1.3%	2.4%	1.4%
	real_video	3.5%	5.3%	3.6%
RT-DETR-x	real_clean	30.1%	50.2%	34.8%
	real_short	3.5%	5.8%	4.1%
	real_video	4.6%	7.3%	5.5%
Faster R-CNN	real_clean	21.3%	39.2%	28.7%
	real_short	0.8%	1.6%	0.9%
	real_video	1.5%	3.0%	1.8%
RetinaNet	real_clean	22.6%	39.4%	27.0%
	real_short	0.9%	1.8%	1.0%
	real_video	1.9%	3.1%	2.0%

TABLE 9. Real-Capture MOT17-09-FRCNN Still-Frame Tracking (ByteTrack-Style Greedy Association, 450 Frames)

Detector	Condition	HOTA	IDF1
YOLO26x	real_clean	15.8%	12.1%
	real_short	8.7%	6.5%
	real_video	10.0%	8.7%
RT-DETR-x	real_clean	14.2%	10.4%
	real_short	8.1%	5.2%
	real_video	9.6%	6.8%
Faster R-CNN	real_clean	14.4%	9.6%
	real_short	6.8%	6.0%
	real_video	10.5%	7.1%
RetinaNet	real_clean	14.2%	9.8%
	real_short	5.1%	4.4%
	real_video	6.8%	5.9%

F. SOFTWARE SIMULATION SUPPLEMENTARY EXPERIMENTS

This section uses software simulation to address two questions: (1) what is the machine readability of a single protected subframe obtained from the display compositor? (2) what is the attacker's recovery boundary under zero camera noise with perfect alignment? Question (1) also characterizes single-frame screenshots: a screenshot tool captures the framebuffer at one instant, corresponding to one protected subframe. The 0–2.6% recovery rates below apply to this

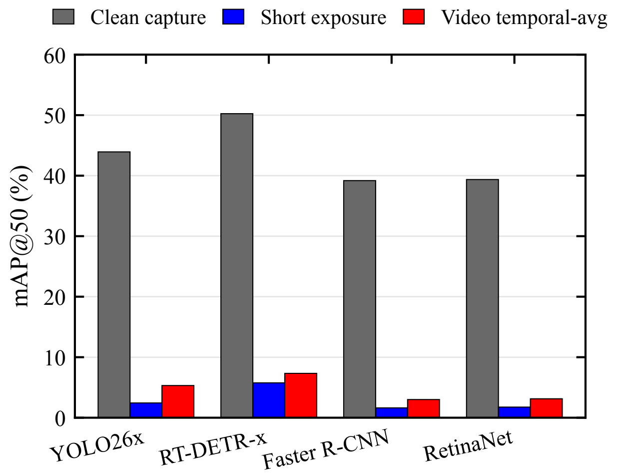

FIGURE 7. Real-capture detection: mAP50 drop for four detectors under protected conditions.

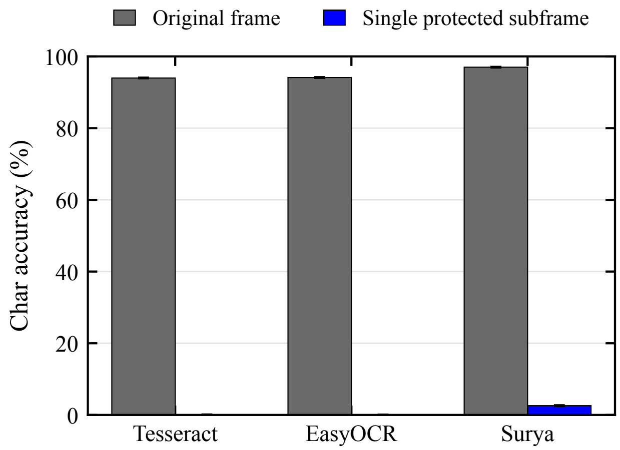

FIGURE 8. Character recovery for three OCR engines on original frames and single protected subframes; error bars are 95% bootstrap intervals resampled by corpus sample.

TABLE 10. Synthetic Single-Subframe OCR: Three-Engine Character Recovery (120 Samples, $n = 4$, $\varepsilon = 8/255$)

OCR engine	Original frame	Single subframe	Reduction
Tesseract	94.0%	0.0%	94.0%
EasyOCR	94.1%	0.0%	94.1%
Surya	97.0%	2.6%	94.4%

scenario. The boundary is clear: if malware intercepts the original frame before masking, or screen recording accumulates a full cycle, the method offers no protection. We claim only single-frame-level mitigation, not a general "anti-screenshot" guarantee.

1) Synthetic Single-Subframe OCR

Using Tesseract [43], EasyOCR [42], and Surya [41] on 120 synthetic samples, character recovery drops from 94.0–97.0% to 0–2.6% across three engines (Table 10). The result covers the current corpus, parameters, and engine set; stratified breakdowns can check category trends but are insufficient to prove effectiveness across all languages, layouts, or recognizers.

Fig. 8 visualizes these results. Single-subframe recovery does not exceed 2.6% for any engine, confirming the finding is not driven by one recognizer. Broader generalization requires additional engines and corpora.

2) Theoretical Security Boundary

When an attacker under ideal conditions accumulates $\geq n$ frames covering a full cycle, the complementarity is exactly exploited and adversarial noise is simultaneously cancelled per (1), recovering 95.2% (per-sample strongest attack). This is the *fundamental boundary* of any linear temporal scheme. The capture-hardened profile reduces recovery under the current real-capture integration conditions, but video averaging still yields 47.9% character recovery; it can therefore only be called mitigation, not closure of this boundary.

Information-theoretic perspective. Under the ideal single-subframe model, exactly $1/n$ of pixels are illuminated per frame: at n=4, 25% of image pixels are retained. The normalized mutual information proxy $I(X;Y)/H(X)$ measured in the Pareto sweep is 0.377 for n=4 at 240 Hz, higher than the naïve $1/n$=0.25 pixel-retention fraction because text pixels exhibit strong spatial correlation (neighboring pixels in a glyph co-predict each other). This gap between pixel-level retention (25%) and information-level retention (37.7%) explains why character recovery does not simply scale as $1/n$: sparse but spatially correlated fragments preserve more recognizable structure than a random 25% sample of an i.i.d. image would. The actual real-capture mask-only recovery of 19.2% falls below even the $1/n$ pixel fraction, because additional ISP degradation, moiré, and optical blur destroy the spatial correlation the recognizer would need to exploit. A formal closed-form bound on character recovery as a function of n would require modeling the joint distribution of character topology and mask geometry, which is beyond the scope of this feasibility study.

3) Comparison with Software-Proxy Baselines

In a software simulation on 102 corpus samples (expanded from an earlier 9-sample stratified pilot), character recovery for dimming, Gaussian blur, pixelation, off-axis privacy-filter proxy, and a single temporal-mask subframe is 94.5%, 71.7%, 0.8%, 94.2%, and 0%, respectively. Dimming by 50% matches the unprotected condition at 94.5%, providing no OCR reduction; Gaussian blur and the off-axis proxy both retain >70% recovery. Parameters, physical conditions, and user-experience costs are not equivalently matched; the "privacy filter" is a software off-axis proxy, not a physical product test. Fig. 9 shows the trade-off under these proxy settings and does not establish superiority over all passive approaches.

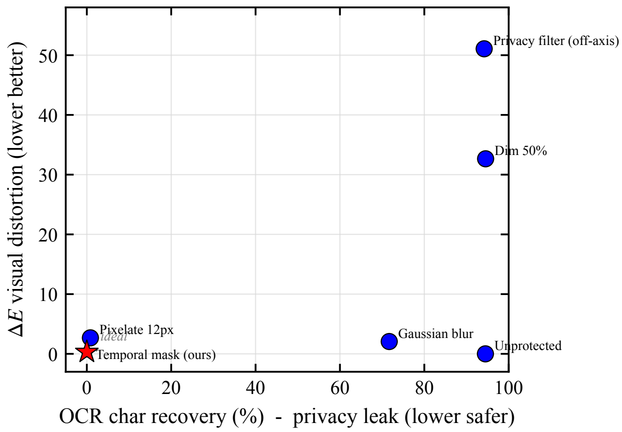

FIGURE 9. Software-proxy baselines on 102 corpus samples: horizontal axis is OCR character recovery, vertical axis is digital reconstruction ´E_{00}. Method parameters and physical conditions are not equivalently matched; the figure serves only as an exploratory comparison.

TABLE 11. COCO val2017 Simulated Detection Pipeline Diagnostics (4 Detectors, 8 Images; Not a Dataset-Level Performance Estimate)

Detector	Condition	mAP	mAP50	AR
YOLO26x	clean	48.1%	58.3%	51.0%
	single subframe	0.0%	0.0%	0.0%
	temporal avg.	14.3%	18.3%	14.3%
RT-DETR-x	clean	47.1%	58.7%	53.4%
	single subframe	5.2%	6.0%	5.2%
	temporal avg.	16.8%	23.7%	17.9%
Faster R-CNN	clean	38.2%	52.7%	43.6%
	single subframe	3.0%	4.4%	3.0%
	temporal avg.	11.1%	16.5%	11.1%
RetinaNet	clean	33.2%	47.1%	35.3%
	single subframe	3.3%	4.2%	3.3%
	temporal avg.	9.6%	13.2%	9.6%

The stripe/glyph parameter trade-off for the anti-OCR profile is shown in Fig. 10.

4) Detection and Tracking Simulation

Supplementary simulations were run on COCO val2017 [38] and MOT17 [39]. The four detectors are YOLO26x [48], RT-DETR-x [49], Faster R-CNN [50], and RetinaNet [51]. The COCO simulation used only 8 images, positioned as implementation pipeline diagnostics (Table 11). The MOT simulation covers 7 sequences totaling 5,316 frames, using a ByteTrack-inspired [52] greedy association fallback implementation (Table 12). Because TrackEval failed to run, MOTA/MOTP/IDF1 in the table come from an in-project approximate evaluation backend and are not equivalent to official ByteTrack/TrackEval results.

On these 8 images, single-subframe mAP50 across four detectors is 0–6.0%. This sample size is sufficient only to check trends, not to estimate COCO population performance. In the 5,316-frame approximate tracking results, single-subframe

TABLE 12. MOT17 Simulated Tracking (ByteTrack-Style Greedy Association; In-Project Approximate Metrics; 5,316 Frames)

Detector	Condition	MOTA	MOTP	IDF1
YOLO26x	clean	31.1%	81.7%	27.1%
	single subframe	0.1%	74.8%	0.1%
	temporal avg.	9.1%	78.1%	12.0%
RT-DETR-x	clean	24.1%	79.8%	38.2%
	single subframe	−2.3%	73.2%	5.4%
	temporal avg.	−11.4%	75.0%	14.7%
Faster R-CNN	clean	3.8%	77.9%	35.6%
	single subframe	−2.5%	71.2%	4.5%
	temporal avg.	−2.7%	73.2%	14.0%
RetinaNet	clean	−5.4%	77.1%	30.5%
	single subframe	0.4%	70.6%	2.7%
	temporal avg.	−2.9%	73.8%	11.9%

IDF1 is 0.1–5.4%, though MOTA exhibits negative values and nonmonotonic behavior. Fig. 11 summarizes residual capability of a single protected subframe relative to the clean baseline across the current evaluation; it does not represent untested models.

5) Security–Usability Trade-off

The model sweep covers $n \in \{2, 4, 6, 8\}$ and refresh rates $\in \{144, 240, 360\}$ Hz across 12 combinations. The Pareto front is defined on the (FPI, normalized mutual information) bi-objective minimization. All 12 configurations are retained on the front without exclusion; FPI values are annotated per point for reference rather than used as a hard threshold, because the FPI formula has a known discontinuity at f_p=60 Hz (§V-A) and is not derived from IEEE Std 1789-2015 [40] or any clinical flicker standard. Representative configurations include $n = 8$ @ 360 Hz (FPI = 0.219, normalized mutual information proxy = 0.180), which achieves the lowest single-subframe information retention but the highest predicted flicker risk; $n = 6$ @ 360 Hz (FPI = 0.033, normalized mutual information proxy = 0.249, $\Delta E_{00} = 0.60$, SSIM = 0.9995); and the base configuration $n = 4$ @ 240 Hz (FPI = 0.030, normalized mutual information proxy = 0.377, $\Delta E_{00} = 0.34$). Higher n trades lower information leakage for higher FPI and ΔE_{00}; whether the flicker cost of n=8 is acceptable is a user-study question, not one the current digital model can answer. All candidates are selected by a simplified digital model and have not been validated on physical panels or in user studies; none constitutes a deployment recommendation. Fig. 12 shows the Pareto front across different n and refresh-rate combinations on the two proxy axes.

VI. DISCUSSION

A. EVIDENCE HIERARCHY AND CLAIM SCOPE

The most directly supported conclusion is: within a single-UVC-camera feasibility study, temporal pixel masking substantially reduces conventional OCR character recovery and exact-match rates from short-exposure screen photographs under the tested S600 camera, display, corpus, and 9 geomet-

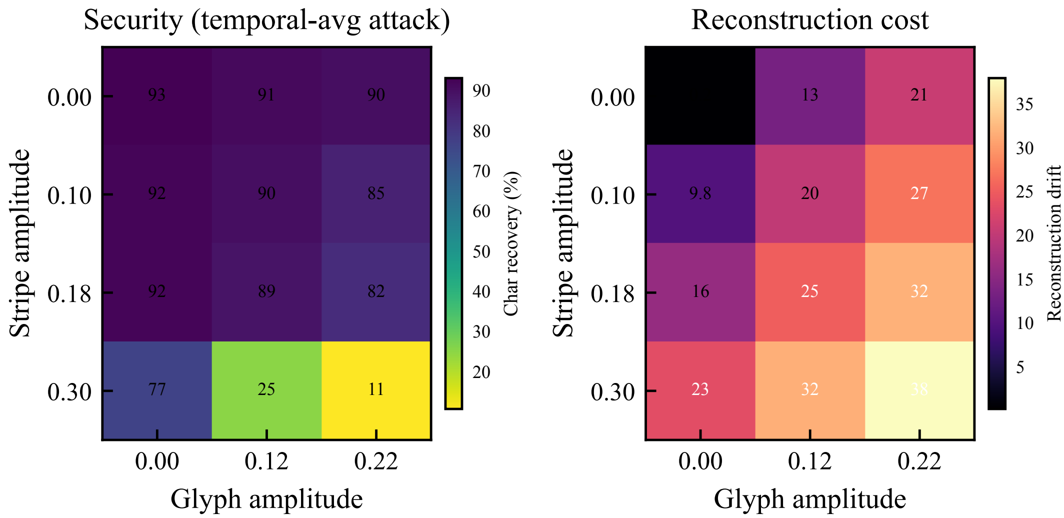

FIGURE 10. **Anti-OCR profile stripe × glyph amplitude grid (temporal-average attack): left panel shows character recovery, right panel shows digital reconstruction drift. High stripe and glyph amplitude regions descriptively reduce character recovery to ∼11% across the current 12 parameter grid points (16 samples per point) while increasing reconstruction cost. No significance testing was performed.**

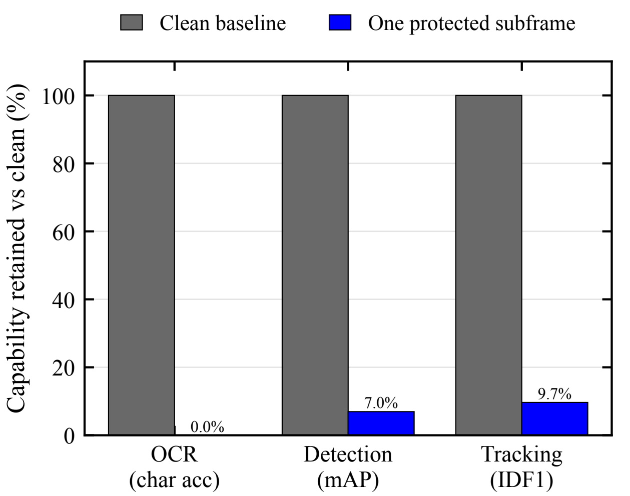

FIGURE 11. **Residual capability of a single protected subframe relative to the clean baseline (clean = 100%) across the current evaluation: OCR, detection, and tracking are approximately 0–9.7%. This figure summarizes only the tested models and metrics.**

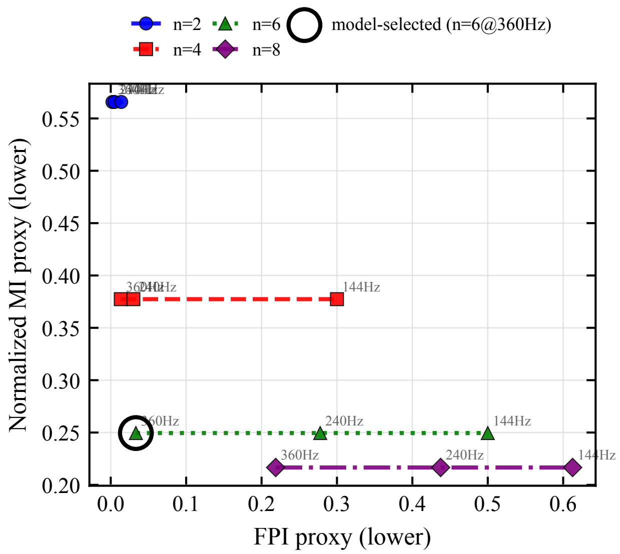

FIGURE 12. **Security–usability proxy Pareto front in the digital model: *n*× refresh rate. Neither FPI nor normalized mutual information is a direct measure of user experience or real attack success rate.**

ric conditions. The 10,575 real-capture images reflect component, parameter, and attack-condition coverage within this UVC link. Because the corpus is unbalanced and covers one webcam, the total sample count is not cross-device or cross-scenario inference evidence. Detection/tracking results, VLM probes, and simulations respectively answer whether the degradation transfers to other visual tasks, whether stronger recognizers break through, and where the full-cycle integration boundary lies, and together they define boundaries rather than extend the primary claim.

B. THE MATHEMATICAL TENSION OF LINEAR INTEGRATION

The scheme exploits the gap between human temporal integration and single-frame camera sampling, but human integration is fundamentally linear summation ($\sum \mathbf{I}_k = \mathbf{I}$); a camera performing the same multi-frame accumulation recovers the original. This is the fundamental boundary of all persistence-of-vision temporal schemes. Research on VLM adversarial robustness [46] further shows that large models extract partial semantic information from perturbed images; the probes in §V-C confirm this: VLMs recover large-font content from single short-exposure photographs and obtain stronger signals from temporal-average and long-exposure views. Full-cycle integration recovers 95.2% (§V-F2).

C. PHYSICAL-LAYER CONFOUNDS AND THE SIMULATION-TO-REAL GAP

Two physical-layer mechanisms systematically separate real-capture results from idealized single-subframe simulation.

Direction 1: real short-exposure recovery exceeds simulation (favoring the attacker). Simulated single-subframe OCR yields 0–2.6% character recovery (Table 10), whereas real-capture short-exposure recovery under comparable masking is 19.2% (mask-only, Table 4). Two factors contribute: (a) *Rolling-shutter row mixing*: the S600's estimated readout time (10–16 ms at 1080p/60 fps) far exceeds the 4.17 ms subframe duration, so different rows of a single photograph sample different subframes, increasing the per-frame pixel coverage beyond $1/n$. (b) *LCD pixel response ghosting*: the AOC 24G51Z's 1 ms GtG (manufacturer spec) occupies $\sim 24\%$ of one subframe slot; slower gray-level transitions blend adjacent subframes temporally, leaking additional information into a single exposure. Both effects are already embedded in the real-capture measurements; the simulation numbers represent an idealized lower bound that no rolling-shutter camera on this display can reach.

Direction 2: real integration recovery is lower than simulation (favoring the defender). Ideal full-cycle digital integration recovers 95.2% (§V-F2), whereas real-capture video temporal averaging on the deployed profile yields 71.1% and capture-hardened yields 47.9%. Camera ISP nonlinearities (gamma, white balance, noise reduction), moiré interference between the display pixel grid and the sensor Bayer pattern, optical blur from defocus and lens aberration, and imperfect temporal registration across frames all degrade the attacker's integration quality. This gap suggests that the idealized 95.2% ceiling overestimates the practical integration threat from a single consumer camera, though a purpose-built attack setup could narrow it.

These two directions imply that the simulation and real-capture experiments answer different questions: simulated single-subframe recovery characterizes the algorithm's theoretical information hiding, while real-capture recovery characterizes the end-to-end system including panel, camera, and ISP confounds.

D. ROLE AND COST OF THE CAPTURE-HARDENED PROFILE

The capture-hardened profile breaks strict completeness through stripe and glyph perturbations, reducing integration recovery at the cost of worsened reconstruction metrics. In the digital pipeline, its full-cycle SSIM is 0.763 and ΔE_{00} is 4.62, far below the deployed profile's 0.948 and 1.46, respectively. It achieves 0% short-exposure exact-match and 5.6% video temporal-average exact-match (§V-B), but character recovery remains at 47.9%, which is mitigation rather than blocking. Whether users accept the visible stripes and grain remains untested: the study in §V-D covers only the deployed profile and core mask conditions.

E. VLM ATTACKERS AND COUNTERMEASURE DIRECTIONS

Protection failure concentrates on large-font short snippets (§V-C), and temporal averaging further amplifies the signal: glyph contour fragments in a single subframe suffice for language-prior completion, and multi-cycle coverage approaches the original. We retain this negative result as a primary boundary contribution because it directly constrains the method's deployability: the scheme mitigates conventional OCR short-exposure capture but does not resist modern VLM screen-reading. Potential countermeasure directions include: (1) **font-size-adaptive mask granularity**, scaling the mask cell with stroke width so that a single subframe no longer preserves character topology; (2) **complementary glyph decoys**, injecting false strokes within the $\sum_k \mathbf{N}_k = \mathbf{0}$ framework that cancel over a cycle, causing single-frame attackers to obtain confident but *incorrect* text rather than partial truth, directly countering prior-based completion; (3) **field-level content policies**, rendering high-risk fields in dense small font or enabling the hardened profile per field. None have been experimentally validated. Adversarial perturbations targeting VLM encoders are further constrained by digital-to-physical transferability (§II). More generally, any information visible to the observer within the integration window is in principle recoverable by a full-cycle camera paired with a strong-prior model; VLMs merely reduce the fragment count needed. The defense objective should shift from "unreadable to any attacker" toward increasing attack cost and preventing silent bulk collection.

F. ADAPTIVE VIDEO ATTACKER

The threat model assumes the attacker does not know the current mask seed or phase, but an adaptive attacker recording video can infer the cycle frequency without either. At 60 fps capture against a 60 Hz display cycle (basic n=4 at 240 Hz), each video frame samples approximately one subframe slot; accumulating $\geq n$ consecutive frames, a minimum of $n/60 \approx$ 67 ms of video, probabilistically covers all slots. The attacker need not identify the phase starting point: exhaustive search over n candidate phase offsets and selection of the alignment yielding maximum reconstruction quality is computationally trivial (n=4 candidates). This analysis confirms that the video temporal-averaging attack already reported in Table 4 (71.1% char for deployed, 47.9% for capture-hardened) represents a realistic adaptive attacker lower bound rather than a theoretical worst case. The practical barrier to this attack is not algorithmic but operational: the attacker must hold a camera on the target screen for at least one display cycle ($\sim$17–21 ms), which is much easier than the short-exposure "grab and go" scenario but still requires deliberate sustained aiming.

G. DEPLOYMENT PATH

Real-world deployment would require OS driver-level frame-buffer injection (DXGI / KMS-DRM / CoreDisplay), panel/backlight timing coordination, and high-refresh panels meeting the target cycle rate. These steps introduce color management, luminance saturation, frame drops, and synchronization errors that could alter algorithm behavior; they cannot be treated as engineering packaging that leaves conclusions unchanged. The deployed profile's 96.5% sensitive-token recovery under long exposure means credentials and similar high-risk fields should not rely on it alone. Such fields should use the capture-hardened profile or dense small-font rendering under a field-level policy, with the caveat that these options impose unresolved readability costs and do not close the video/VLM boundary.

H. ETHICS STATEMENT

This research is defensive in nature, aiming to protect screen content from unauthorized photography. The attack analysis (§V-F2) is intended for honest assessment of security boundaries, not as an attack guide. The planned user study would involve human subjects in minimal-risk research; its protocol specifies informed consent, photosensitivity screening, and voluntary withdrawal as described in §V-D.

I. LIMITATIONS

1) The user study (§V-D) is a protocol, not completed evidence: no participant data exist yet, so the deployed profile's usability remains unvalidated. Even when conducted in a controlled 240 Hz laboratory, external validity will require broader participant samples and display conditions.
2) Immediate visual comfort and subjective anti-peeking effectiveness are single-item scales. The former does not extend to long-term fatigue; the latter estimates bystander legibility without independent observers or controlled viewing geometry and is not objective shoulder-surfing evidence.
3) The PoC is application-layer only, without OS drivers or hardware backlight. Luminance, HDR, and color-difference conclusions come from digital models, lacking photometric or panel-response measurements. OS compositor jitter, VSync mismatch, and dropped frames can shift subframe boundaries unpredictably; if a short-exposure window straddles the intended slot edge, the camera captures cross-slot content and real recovery may exceed the values reported here.
4) Real-capture OCR used one S600 UVC webcam; ambient illuminance was not recorded; exposure values have two calibration sets. The unbalanced design with repeated content means bootstrap intervals do not substitute for cross-device replication. Uncontrolled ambient light variation may affect results through two paths: (a) the camera's auto-exposure, auto-white-balance, and auto-gain algorithms respond to ambient conditions, altering effective exposure on the screen content; (b) ambient light reflected off the screen surface changes the captured signal-to-noise ratio. If geometric configurations were captured at different times of day, systematic ambient variation could confound geometric-factor analysis.
5) Real-capture detection used 150 COCO images at a single position; tracking used 450 still-frame captures from one MOT17 sequence, not continuous video. Simulated detection used 8 images with approximate metrics.
6) VLM evaluation covers one camera, 0° on-axis, one profile, and three commercial API models (12 video segments per cell). Models may update over time. Public document content may inflate recovery via training-data memorization. We did not evaluate multi-cycle adaptive reconstruction, rolling-shutter phase search, or pre-masking malware interception. Full-cycle integration remains the fundamental boundary.
7) All 12 test content items are English/ASCII text. CJK characters (Chinese, Japanese, Korean) have much higher stroke density than Latin letters; at the same mask granularity, CJK strokes are more likely to be disrupted (stroke widths closer to mask cell size), but higher inter-character visual similarity may also increase OCR confusion. The net effect on protection efficacy for non-Latin scripts is unknown and left to future work.

VII. CONCLUSION

We implemented and evaluated a temporal pixel masking method that exploits integration-time differences between human vision and short-exposure cameras. In a 10,575-image feasibility study on a single UVC webcam, the method reduces best-of-engine OCR character recovery from 94.1% to 15.1% (deployed) and 5.0% (capture-hardened) under calibrated short exposure, with exact-match falling to 0.4% and 0%. Four object detectors' mAP50 drops from 39.2–50.2% to 1.6–5.8%.

The security boundary is sharply defined. Video temporal averaging ($\geq$67 ms) recovers 47.9–71.1% character; under long exposure, temporal masking can paradoxically assist the attacker by bringing effective luminance into the sensor's dynamic range (deployed 60.9% versus unprotected 47.3%). VLM probes show that modern models recover up to 77.8% exact-match on large-font short snippets at 0.5 m, bypassing the binarization bottleneck that temporal masking disrupts. These boundaries confine the contribution to conventional OCR short-exposure mitigation and provide empirical design constraints (content type, font size, and attacker capability) for next-generation display-privacy research.

Real-world usability validation, cross-device generalization, and VLM-aware countermeasures remain open.

DATA AND CODE AVAILABILITY

The software implementation, experimental configurations, per-sample OCR results, and VLM probe prompts, parameters, and per-call outputs are planned for release as a de-identified research archive. The legacy `vlm` profile tag in historical records denotes the capture-hardened configuration (§IV-E). If the user study is conducted, its database `study_formal.db` (containing student IDs and names) will not be released; public materials will include only de-identified results, exclusion audits, and analysis scripts. **[TODO: Final archive URL, version number, and persistent identifier to be added.]**

ACKNOWLEDGMENT

[TODO: Acknowledgment to be filled.]

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